

Branch: M. Pharm. (Pharmacology)
Subject: Principles of Drug Discovery
Date: 19/06/2024

Time: 3 hrs

Semester: II
Subject Code: MPL203T
Marks: 75

Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Attempt any FIVE questions of the following: (10 Marks)

- A] Define target identification and target validation.
- B] Write functions of Zinc finger proteins.
- C] Define regression analysis and state its importance.
- D] Enlist different physicochemical parameters considered in QSAR.
- E] Define siRNA and give its advantages.
- F] Give importance of drug likeness screening in rational drug design.
- G] Define “Prodrug” and give one example.

Q. No.2 Attempt any FOUR questions of the following: (20 Marks)

- A] Describe different statistical methods used in QSAR analysis.
- B] Explain the role of transgenic animals in target validation.
- C] Write a note on Free Wilson and Hansch analysis.
- D] Elaborate the concept of pharmacophore mapping and pharmacophore based screening.
- E] Write a note on Prodrug design.
- F] Describe HTS and its role in rational drug design.

Q. No.3 Attempt any THREE questions of the following: (45 Marks)

- A] Write elaborately on the role of genomics, proteomics and bioinformatics in target discovery and validation process. Explain the importance of microarrays.
- B] Differentiate between SAR and QSAR. Discuss the history, development, advantages and disadvantages of QSAR.
- C] Write in detail about the protein structure and methods used for the computational prediction of protein structure.
- D] Explain the concept of molecular docking, discuss its types and elaborate the docking based screening.
- E] Write a note on following: (Any THREE)
 - (1) NMR (2) Partial least square analysis (PLS) (3) De novo drug design
 - (4) CoMFA (5) Combinatorial Chemistry

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